

Assignment 1

a.

Aim. Write a program in C for matrix subtraction.
Show the statement level calculation of time complexity
in the right column. Write the expression for
order of time complexity

Time complexity

```
#include <stdio.h>
```

```
void main() {
```

```
    int a[3][3];
```

1

```
    int b[3][3];
```

1

```
    printf("A array");
```

C₁

```
    for(int i=0; i<3; i++){
```

1 + 3 × 2

```
        for(int j=0; j<3; j++){
```

3 × (1 + 3 × 2)

```
            scanf("%d", &a[i][j]);
```

(C₂ + 3) × 3 × 3

```
        }
```

```
    }
```

```
    printf("B array");
```

C₁

```
    for(int i=0; i<3; i++){
```

1 + 3 × 2

```
        for(int j=0; j<3; j++){
```

3 × (1 + 3 × 2)

```
            scanf("%d", &b[i][j]);
```

(C₂ + 3) × 3 × 3

```
        }
```

```
    }
```

```
    for(int i=0; i<3; i++){
```

1 + 3 × 2

```
        for(int j=0; j<3; j++){
```

3 × (1 + 3 × 2)

```
            a[i][j] = a[i][j] - b[i][j];
```

(9 + 2) × 9

```
        }
```

```
    }
```

```

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        printf("%d", a[i][j]);
    }
    printf("\n");
}

```

$1 + 2 \times 3$
 $3(1 + 2 \times 3)$
 $(c_1 + 3) \times 9$
 $c_1 \times 3$

Time complexity for 3×3 matrix
 $= 12c_1 + 18c_2 + 296$

For $n \times n$ matrix

Time complexity = $1 + 1 + c_1 + 1 + n \times 2 + n(1 + 2n) +$
 $(c_2 + 3)n^2 + c_1 + 1 + 2n + n(1 + 2n) + (c_2 + 3)n^2$
 $+ 1 + 2n + n(1 + 2n) + n^2(11)$
 $+ 1 + 2n + n(1 + 2n) + (c_1 + 3)n^2 + nc_1$

$$= \cancel{6} + 2c_1 + nc_1 + \cancel{6n^2} + 3n^2$$

The order of this expression is $O(n^2)$

Assignment 1

b.

Aim: Write a program in C to show the region of memory used by code, data, stack and extra segments. Calculate the space required for the program. Write the expression for order of space complexity.

```
#include <stdio.h>
void f(int a){
    printf("In Local variable of caller f has value = %d\n", a, &a);
    if(--a){
        f(a);
    }
}
void main(){
    int z = 4;
    printf("In Local variable z of caller function\n", z, &z);
    f(z);
}
```

function	variable	Value	address
main	a	4	000000000 61FE4C
f	a	4	000000000 61FE20
f	a	3	000000000 61FDFO
f	a	2	61FD00
f	a	1	61FD90

Hex byte difference = 30, 48 bytes

extra segments used & data segment size is zero
space used by code ^{segment} = ~~82 KB~~ 13.8 KB

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Array A2
3
4
5
6
7
8
9
10
Array B1
2
3
4
5
6
7
8
9
111
111
111
❖ tarun@Tarun-PC: ~/Downloads\$

☆



A2
Executable
13.8 kB

Parent Folder
/home/tarun/Downloads/

Accessed
7/23/2026, 6:11:22 PM

Modified
7/23/2026, 6:11:22 PM

Created
7/23/2026, 6:11:22 PM

Permissions
Read and Write >

Executable as Program ☒

```
Local variable z of caller function main has value=4 and address=0x7fffffff94c  
  
Local Variable of a calle f has value=4 and address=0x7fffffff92c  
  
Local Variable of a calle f has value=3 and address=0x7fffffff90c  
  
Local Variable of a calle f has value=2 and address=0x7fffffff8ec  
  
Local Variable of a calle f has value=1 and address=0x7fffffff8cc  
❖ tarun@Tarun-PC: ~/Downloads$
```